United States Patent Application Publication	20250286515
Kind Code	A1
Publication Date	September 11, 2025
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POWER AMPLIFIER CIRCUIT

Abstract

A power amplifier circuit includes a carrier amplifier, which amplifies a first signal and outputs a first amplified signal from a first output terminal, a peak amplifier, which amplifies a second signal and outputs a second amplified signal from a second output terminal, a first inductor, connected to the first output terminal and a first node, a first capacitor, connected to the first node and the ground, a second inductor, connected to the second output terminal and a second node, a second capacitor, connected to the second node and the ground, a third inductor, connected to the first node and a third node connected to a load, and a third capacitor, connected to the second node and the third node.

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Family ID: **91196210**

Appl. No.: **19/215414**

Filed: **May 22, 2025**

Foreign Application Priority Data

JP	2022-188244	Nov. 25, 2022

Related U.S. Application Data

parent WO continuation PCT/JP2023/042256 20231124 PENDING child US 19215414

Publication Classification

Int. Cl.: **H03F1/02** (20060101); **H03F3/21** (20060101)

U.S. Cl.:

CPC **H03F1/0288** (20130101); **H03F3/211** (20130101); H03F2200/48 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This is a continuation of International Application No. PCT/JP2023/042256 filed on Nov. 24, 2023 which claims priority from Japanese Patent Application No. 2022-188244 filed on Nov. 25, 2022. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND ART
[0002] The present disclosure relates to a power amplifier circuit.
[0003] A Doherty amplifier includes a carrier amplifier and a peak amplifier (for example, see Patent Document 1). A semiconductor power amplifier includes a balanced amplifier (for example, see Patent Document 2). [0004] Patent Document 1: Japanese Unexamined Patent Application Publication No. 2012-29239 [0005] Patent Document 2: Japanese Unexamined Patent Application Publication No. 2014-225791
BRIEF SUMMARY
[0006] The Doherty amplifier described in Patent Document 1 achieves high efficiency, but needs a quarter-wave line, resulting in a large space required. In contrast, the balanced amplifier described in Patent Document 2 achieves resistance to fluctuations in load impedance, but has degraded efficiency. That is, it is difficult to obtain a power amplifier circuit which achieves both high efficiency and high resistance to fluctuations in load impedance.
[0007] The present disclosure is made in view of such a situation, and an object thereof is to provide a power amplifier circuit which achieves a reduction in size, high efficiency, and high resistance to fluctuations in load impedance.
[0008] A power amplifier circuit according to an aspect of the present disclosure includes a carrier amplifier, a peak amplifier, a first inductor, a first capacitor, a second inductor, a second capacitor, a third inductor, and a third capacitor. The carrier amplifier amplifies a first signal, and outputs a first amplified signal from a first output terminal. The peak amplifier amplifies a second signal different in phase from the first signal, and outputs a second amplified signal from a second output terminal. The first inductor has a first end connected to the first output terminal, and a second end connected to a first node. The first capacitor has a first end connected to the first node, and a second end connected to the ground. The second inductor has a first end connected to the second output terminal, and a second end connected to a second node. The second capacitor has a first end connected to the second node, and a second end connected to the ground. The third inductor has a first end connected to the first node, and a second end connected to a third node connected to a load. The third capacitor has a first end connected to the second node, and a second end connected to the third node.
[0009] A power amplifier circuit according to another aspect of the present disclosure includes a carrier amplifier, a peak amplifier, a first inductor, a first capacitor, a second inductor, a second capacitor, and a 90° hybrid coupler. The carrier amplifier amplifies a first signal, and outputs a first amplified signal from a first output terminal. The peak amplifier amplifies a second signal different in phase from the first signal, and outputs a second amplified signal from a second output terminal. The first inductor has a first end connected to the first output terminal, and a second end connected to a first node. The first capacitor has a first end connected to the first node, and a second end

connected to the ground. The second inductor has a first end connected to the second output terminal, and a second end connected to a second node. The second capacitor has a first end connected to the second node, and a second end connected to the ground. The 90° hybrid coupler has a first input port, a second input port, an output port, and an isolation port. The first input port is connected to the first node and is supplied with the first amplified signal. The second input port is connected to the second node and is supplied with the second amplified signal. The output port is connected to a third node connected to a load and is a port from which an output signal is outputted.

[0010] A power amplifier circuit according to another aspect of the present disclosure includes a carrier amplifier, a peak amplifier, a first capacitor, a first inductor, a second capacitor, a second inductor, a third capacitor, and a third inductor. The carrier amplifier amplifies a first signal, and outputs a first amplified signal from a first output terminal. The peak amplifier amplifies a second signal different in phase from the first signal, and outputs a second amplified signal from a second output terminal. The first capacitor has a first end connected to the first output terminal, and a second end connected to a first node. The first inductor has a first end connected to the first node, and a second end connected to the ground. The second capacitor has a first end connected to the second output terminal, and a second end connected to a second node. The second inductor has a first end connected to the second node, and a second end connected to the ground. The third capacitor has a first end connected to the first node, and a second end connected to a third node connected to a load. The third inductor has a first end connected to the second node, and a second end connected to the third node.

[0011] A power amplifier circuit according to another aspect of the present disclosure includes a carrier amplifier, a peak amplifier, a first capacitor, a first inductor, a second capacitor, a second inductor, and a 90° hybrid coupler. The carrier amplifier amplifies a first signal, and outputs a first amplified signal from a first output terminal. The peak amplifier amplifies a second signal different in phase from the first signal, and outputs a second amplified signal from a second output terminal. The first capacitor has a first end connected to the first output terminal, and a second end connected to a first node. The first inductor has a first end connected to the first node, and a second end connected to the ground. The second capacitor has a first end connected to the second output terminal, and a second end connected to a second node. The second inductor has a first end connected to the second node, and a second end connected to the ground. The 90° hybrid coupler has a first input port, a second input port, an output port, and an isolation port. The first input port is connected to the first node and is supplied with the first amplified signal. The second input port is connected to the second node and is supplied with the second amplified signal. The output port is connected to a third node connected to a load and is a port from which an output signal is outputted.

[0012] A power amplifier circuit according to another aspect of the present disclosure includes a carrier amplifier, a peak amplifier, a first inductor, a first capacitor, a second inductor, a third capacitor, a fifth inductor, and a second capacitor. The carrier amplifier amplifies a first signal, and outputs a first amplified signal from a first output terminal. The peak amplifier amplifies a second signal different in phase from the first signal, and outputs a second amplified signal from a second output terminal.

[0013] The first inductor has a first end connected to the first output terminal, and a second end connected to a first node connected to a load. The first capacitor has a first end connected to the first node, and a second end connected to the ground. The second inductor has a first end connected to the second output terminal, and a second end connected to a second node. The third capacitor has a first end connected to the second node, and a second end connected to the first node. The fifth inductor and the second capacitor are connected in series between the second node and the ground.

Effects of Disclosure

[0014] The present disclosure may provide a power amplifier circuit which achieves a reduction in size, high efficiency, and high resistance to fluctuations in load impedance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. **1** is a circuit diagram of a power amplifier circuit **101**.

[0016] FIG. **2** is a circuit diagram of a power amplifier circuit **900** which is a reference example.

[0017] FIG. **3** is a diagram illustrating a simulation result of frequency characteristics of the change in phase across output terminals in the power amplifier circuit **900**.

[0018] FIG. **4** is a diagram illustrating a simulation result of frequency characteristics of the change in phase across output terminals in the power amplifier circuit **101**.

[0019] FIG. **5** is a diagram illustrating a simulation result of frequency characteristics of the phase difference between signals in the power amplifier circuit **900**.

[0020] FIG. **6** is a diagram illustrating a simulation result of frequency characteristics of the phase difference between signals in the power amplifier circuit **101**.

[0021] FIG. **7** is a circuit diagram of a power amplifier circuit **901** which is a reference example.

[0022] FIG. **8** is a diagram illustrating a simulation result of gain characteristics in the power amplifier circuit **101**.

[0023] FIG. **9** is a diagram illustrating a simulation result of gain characteristics in the power amplifier circuit **900**.

[0024] FIG. **10** is a diagram illustrating a simulation result of gain characteristics in the power amplifier circuit **901**.

[0025] FIG. **11** is a diagram illustrating a simulation result of efficiency characteristics in the power amplifier circuit **101**.

[0026] FIG. **12** is a diagram illustrating a simulation result of efficiency characteristics in the power amplifier circuit **900**.

[0027] FIG. **13** is a diagram illustrating a simulation result of efficiency characteristics in the power amplifier circuit **901**.

[0028] FIG. **14** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to output power variations in the power amplifier circuit **101**.

[0029] FIG. **15** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to output power variations in the power amplifier circuit **900**.

[0030] FIG. **16** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to output power variations in the power amplifier circuit **901**.

[0031] FIG. **17** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to output power variations in the power amplifier circuit **101**.

[0032] FIG. **18** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to output power variations in the power amplifier circuit **900**.

[0033] FIG. **19** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to output power variations in the power amplifier circuit **901**.

[0034] FIG. **20** is a diagram illustrating a Smith chart showing a simulation result of a change in impedance Zc relative to a change of load phase in the power amplifier circuit **101**.

[0035] FIG. **21** is a diagram illustrating a Smith chart showing a simulation result of a change in impedance Zp relative to a change of load phase in the power amplifier circuit **101**.

[0036] FIG. **22** is a diagram illustrating exemplary load fluctuation characteristics of power-added efficiency in the power amplifier circuit **101**.

[0037] FIG. **23** is a circuit diagram of a power amplifier circuit **111**.

[0038] FIG. **24** is a circuit diagram of a power amplifier circuit **102**.

[0039] FIG. **25** is a diagram illustrating a simulation result of frequency characteristics of the change in phase across output terminals in the power amplifier circuit **102**.
[0040] FIG. **26** is a diagram illustrating a simulation result of frequency characteristics of the phase difference between signals in the power amplifier circuit **102**.
[0041] FIG. **27** is a circuit diagram of a power amplifier circuit **902** which is a reference example.
[0042] FIG. **28** is a diagram illustrating a simulation result of gain characteristics in the power amplifier circuit **102**.
[0043] FIG. **29** is a diagram illustrating a simulation result of gain characteristics in the power amplifier circuit **902**.
[0044] FIG. **30** is a diagram illustrating a simulation result of efficiency characteristics in the power amplifier circuit **102**.
[0045] FIG. **31** is a diagram illustrating a simulation result of efficiency characteristics in the power amplifier circuit **902**.
[0046] FIG. **32** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to output power variations in the power amplifier circuit **102**.
[0047] FIG. **33** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to output power variations in the power amplifier circuit **902**.
[0048] FIG. **34** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to output power variations in the power amplifier circuit **102**.
[0049] FIG. **35** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to output power variations in the power amplifier circuit **902**.
[0050] FIG. **36** is a circuit diagram of a power amplifier circuit **112**.
[0051] FIG. **37** is a circuit diagram of a power amplifier circuit **122**.
[0052] FIG. **38** is a circuit diagram of a power amplifier circuit **132**.
[0053] FIG. **39** is a diagram illustrating exemplary load fluctuation characteristics of power-added efficiency in the power amplifier circuit **132**.
[0054] FIG. **40** is a circuit diagram of a power amplifier circuit **141**.
[0055] FIG. **41** is a diagram illustrating an exemplary capacitor **202***v*.
[0056] FIG. **42** is a diagram illustrating an exemplary capacitor **183***v*.
[0057] FIG. **43** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zp relative to frequency variations in the power amplifier circuit **141**.
[0058] FIG. **44** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance Zc relative to frequency variations in the power amplifier circuit **141**.
[0059] FIG. **45** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance ZP**2** relative to frequency variations in the power amplifier circuit **141**.
[0060] FIG. **46** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance ZP**1** relative to frequency variations in the power amplifier circuit **141**.
[0061] FIG. **47** is a diagram illustrating exemplary variations in loss relative to frequency variations in an LC circuit **181**.

DETAILED DESCRIPTION

[0062] Embodiments of the present disclosure will be described below in detail by referring to the drawings. The same components are designated with the same reference numerals, and repeated description will be avoided as much as possible.

First Embodiment

[0063] A power amplifier circuit **101** according to a first embodiment will be described. FIG. **1** is a circuit diagram of the power amplifier circuit **101**. As illustrated in FIG. **1**, the power amplifier circuit **101** includes a carrier amplifier **51**, a peak amplifier **52**, LC circuits **61** and **62**, and a 90° hybrid coupler **71**.
[0064] The LC circuit **61** includes a capacitor **201** (first capacitor) and an inductor **211** (first inductor). The LC circuit **62** includes a capacitor **202** (second capacitor) and an inductor **212**

(second inductor).
[0065] The 90° hybrid coupler **71** includes capacitors **203** (third capacitor) and **204** (fourth capacitor), and inductors **213** (third inductor) and **214** (fourth inductor).
[0066] The carrier amplifier **51** amplifies a signal RF**1** (first signal) and outputs an amplified signal RF**3** (first amplified signal) from an output terminal **51***a* (first output terminal).
[0067] The peak amplifier **52** amplifies a signal RF**2** (second signal), which is different in phase from the signal RF**1**, and outputs an amplified signal RF**4** (second amplified signal) from an output terminal **52***a*.
[0068] In detail, the signals RF**1** and RF**2** are, for example, radio frequency (RF) signals. The signals RF**1** and RF**2** are signals generated by a splitter (not illustrated) splitting an input signal. The phases of the signals RF**1** and RF**2** just after being outputted from the splitter are, for example, identical.
[0069] The signal RF**2** is supplied to the peak amplifier **52** through a quarter-wave line (not illustrated), whereas the signal RF**1** is supplied to the carrier amplifier **51** as it is. Therefore, the phase of the signal RF**2** supplied to the peak amplifier **52** is delayed by approximately 90° compared with the phase of the signal RF**1** supplied to the carrier amplifier **51**.
[0070] The output terminal **51***a* of the carrier amplifier **51** is grounded through a capacitor **301**. The output terminal **52***a* of the peak amplifier **52** is grounded through a capacitor **302**. The capacitors **301** and **302** are, for example, parasitic capacities, or are capacities provided for harmonic-wave control.
[0071] The LC circuit **61** is connected between the output terminal **51***a* of the carrier amplifier **51** and an input port Pin**1** in the 90° hybrid coupler **71**. In detail, the inductor **211** (first inductor) in the LC circuit **61** has a first end connected to the output terminal **51***a*, and a second end connected to a node N**1** (first node). The capacitor **201** has a first end connected to the node N**1** and a second end connected to the ground.
[0072] The LC circuit **62** is connected between the output terminal **52***a* of the peak amplifier **52** and an input port Pin**2** in the 90° hybrid coupler **71**. In detail, the inductor **212** in the LC circuit **62** has a first end connected to the output terminal **52***a*, and a second end connected to a node N**2** (second node). The capacitor **202** has a first end connected to the node N**2**, and a second end connected to the ground.
[0073] The inductor **213** in the 90° hybrid coupler **71** has a first end, which is connected to the node N**1** and which is the input port Pin**1** (first input port) supplied with the amplified signal RF**3**, and a second end, which is connected to a node N**3** (third node) and which is an output port Pout. The node N**3** is grounded through a load **86**. An output signal RFout, which is obtained by combining the amplified signals RF**3** and RF**4** with each other, is outputted from the output port Pout through the node N**3** to the load **86**.
[0074] The capacitor **203** has a first end, which is connected to the node N**2** and which is the input port Pin**2** (second input port) supplied with the amplified signal RF**4**, and a second end connected to the node N**3**.
[0075] The capacitor **204** has a first end connected to the node N**1**, and a second end which is an isolation port Piso. The isolation port Piso is not connected to circuit elements external to the 90° hybrid coupler **71**. That is, the isolation port Piso is not connected directly to all the circuit elements, except the capacitor **204** and the inductor **214** which are included in the 90° hybrid coupler **71**.
[0076] The inductor **214** has a first end connected to the node N**2**, and a second end connected to the second end of the capacitor **204**, and is electromagnetically coupled to the inductor **213**.
[0077] FIG. **2** is a circuit diagram of a power amplifier circuit **900** which is a reference example. As illustrated in FIG. **2**, the power amplifier circuit **900** is a Doherty amplifier circuit of the related art. Compared with the power amplifier circuit **101** in FIG. **1**, the power amplifier circuit **900** includes a quarter-wave line **42** instead of the LC circuits **61** and **62**, the 90° hybrid coupler **71**, and the

capacitors **301** and **302**.

[0078] The quarter-wave line **42** has a first end connected to the output terminal **51***a* of the carrier amplifier **51**, and a second end connected to the output terminal **52***a* of the peak amplifier **52**. The node N**3** is connected to the output terminal **52***a*.

[0079] FIG. **3** is a diagram illustrating a simulation result of the frequency characteristics of the change in phase across the output terminals in the power amplifier circuit **900**. The vertical axis indicates the change in phase, whose unit is "°", across the output terminals. The horizontal axis indicates the frequency of the amplified signals RF**3** and RF**4**. The change in phase across the output terminals is a change in phase of the amplified signal RF**3** which is obtained when the amplified signal RF**3** has passed from the output terminal **51***a* of the carrier amplifier **51** through the quarter-wave line **42** to the output terminal **52***a* of the peak amplifier **52**.

[0080] As illustrated in FIG. **3**, in the power amplifier circuit **900**, the change in phase across the output terminals of the amplified signal RF**3** is approximately 90° in the range from 1.68 GHz to 1.98 GHz.

[0081] FIG. **4** is a diagram illustrating a simulation result of the frequency characteristics of the change in phase across the output terminals in the power amplifier circuit **101**. The vertical axis indicates the change in phase, whose unit is "°", across the output terminals. The horizontal axis indicates the frequency of the amplified signals RF**3** and RF**4**. The change in phase across the output terminals in this case is a change in phase of the amplified signal RF**3** which is obtained when the amplified signal RF**3** has passed from the output terminal **51***a* of the carrier amplifier **51** through the nodes N**1** and N**2** to the output terminal **52***a* of the peak amplifier **52**.

[0082] In the power amplifier circuit **101**, the magnitude of the change in phase across the output terminals of the amplified signal RF**3** is greater than or equal to 45° and less than or equal to 135°. In the present embodiment, as illustrated in FIG. **4**, the magnitude of the change in phase across the output terminals of the amplified signal RF**3** is greater than or equal to 60° and less than or equal to 90° in the range from 1.68 GHz to 1.98 GHz. That is, the change in phase across the output terminals, which is comparable to that in the power amplifier circuit **900**, is achieved.

[0083] FIG. **5** is a diagram illustrating a simulation result of the frequency characteristics of the phase difference between signals in the power amplifier circuit **900**. The vertical axis indicates the phase difference, whose unit is "°", between signals. The horizontal axis indicates the frequency of the amplified signals RF**3** and RF**4**. The phase difference between signals indicates the difference between the change in phase, which is obtained when the amplified signal RF**3** outputted from the output terminal **51***a* of the carrier amplifier **51** passes through the quarter-wave line **42** and the load **86**, and the change in phase, which is obtained when the amplified signal RF**4** outputted from the output terminal **52***a* of the peak amplifier **52** passes through the load **86**.

[0084] As illustrated in FIG. **5**, in the power amplifier circuit **900**, the phase difference between the signals is approximately 90° in the range from 1.68 GHz to 1.98 GHz.

[0085] FIG. **6** is a diagram illustrating a simulation result of the frequency characteristics of the phase difference between the signals in the power amplifier circuit **101**. The vertical axis indicates the phase difference, whose unit is "°", between the signals. The horizontal axis indicates the frequency, whose unit is "GHz", of the amplified signals RF**3** and RF**4**. The phase difference between the signals in this case indicates the difference between the change in phase, which is obtained when the amplified signal RF**3** outputted from the output terminal **51***a* of the carrier amplifier **51** passes through the nodes N**1** and N**3** and the load **86**, and the change in phase, which is obtained when the amplified signal RF**4** outputted from the output terminal **52***a* of the peak amplifier **52** passes through the nodes N**2** and N**3** and the load **86**.

[0086] In the power amplifier circuit **101**, the magnitude of the phase difference between the signals is greater than or equal to 60° and less than or equal to 120°. In the present embodiment, as illustrated in FIG. **6**, the magnitude of the phase difference between the signals is approximately 90° in the range from 1.68 GHz to 1.98 GHz. That is, the phase difference between the signals,

which is comparable to that in the power amplifier circuit **900**, is achieved.
[0087] FIG. **7** is a circuit diagram of a power amplifier circuit **901** which is a reference example. As illustrated in FIG. **7**, compared with the power amplifier circuit **101** in FIG. **1**, the power amplifier circuit **901** is a circuit in which the LC circuits **61** and **62** are not disposed.
[0088] FIGS. **8**, **9**, and **10** are diagrams illustrating simulation results of gain characteristics in the power amplifier circuits **101**, **900**, and **901**, respectively. The vertical axis indicates gain, whose unit is “dB”, of a power amplifier circuit. The horizontal axis indicates power, whose unit is “dBm”, of the output signal RFout (hereinafter may be referred to as output power).
[0089] The gain characteristics in FIGS. **8** to **10** are obtained when idle current to the peak amplifier **52** is throttled (hereinafter may referred to as “in Doherty operation”). The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The gain characteristics (see FIG. **8**) in the power amplifier circuit **101** are comparable to the gain characteristics (see FIG. **9**) in the power amplifier circuit **900**.
[0090] FIGS. **11**, **12**, and **13** are diagrams illustrating simulation results of efficiency characteristics in the power amplifier circuits **101**, **900**, and **901**, respectively. The vertical axis indicates the power-added efficiency, whose unit is “%”, of a power amplifier circuit. The horizontal axis indicates output power whose unit is “dBm”.
[0091] The efficiency characteristics in FIGS. **11** to **13** are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The output power indicated by using a dotted line is power which is actually used (hereinafter may be referred to as actual power in use) in the power amplifier circuits **101**, **900**, and **901**.
[0092] The power-added efficiency of the power amplifier circuit **101** at the actual power in use is greater than that of the power amplifier circuit **901** at the actual power in use. The power-added efficiency of the power amplifier circuit **101** at the actual power in use is comparable to that of the power amplifier circuit **900** at the actual power in use.
[0093] FIGS. **14**, **15**, and **16** are diagrams illustrating Smith charts showing simulation results of variations in impedance Zc relative to output power variations in the power amplifier circuits **101**, **900**, and **901**, respectively. Impedance Zc is impedance as seen from the output terminal **51***a* of the carrier amplifier **51** to the load **86**.
[0094] The variations in impedance Zc relative to output power variations, which are illustrated in FIGS. **14** to **16**, are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. When the real number component of impedance Zc is large, the power-added efficiency increases. The direction of the arrow indicated by using a dotted line indicates increase of output power.
[0095] In the power amplifier circuit **901** having only the 90° hybrid coupler **71**, impedance Zc varies along the imaginary number axis (see FIG. **16**). In contrast, in the power amplifier circuit **900**, impedance Zc varies along the real number axis, showing that power-added efficiency is high (see FIG. **15**).
[0096] In the power amplifier circuit **101** in which low-pass filter circuits of the LC circuits **61** and **62** are disposed, variations in impedance Zc relative to output power variations may be made close to that in the power amplifier circuit **900** (see FIGS. **14** and **15**).
[0097] FIGS. **17**, **18**, and **19** are diagrams illustrating Smith charts showing simulation results of variations in impedance Zp relative to output power variations in the power amplifier circuits **101**, **900**, and **901**, respectively. Impedance Zp is impedance as seen from the output terminal **52***a* of the peak amplifier **52** to the load **86**.
[0098] The variations in impedance Zp relative to output power variations in FIGS. **17** to **19** are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. When the real number component of impedance Zp is large, the power-added efficiency increases. The direction of the arrow indicated by using a dotted line indicates increase of output power.
[0099] In the power amplifier circuit **901** having only the 90° hybrid coupler **71**, impedance Zp

varies along the imaginary number axis (see FIG. **19**). In contrast, in the power amplifier circuit **900**, impedance Zp varies along the real number axis, showing that power-added efficiency is high (see FIG. **18**).
[0100] In the power amplifier circuit **101** in which the low-pass filter circuits of the LC circuits **61** and **62** are disposed, variations in impedance Zp relative to output power variations may be made close to that in the power amplifier circuit **900** (see FIGS. **17** and **18**).
[0101] FIGS. **20** and **21** are diagrams illustrating Smith charts showing simulation results of a change in impedance Zc and a change in impedance Zp, respectively, relative to a change of load phase in the power amplifier circuit **101**. FIG. **22** is a diagram illustrating exemplary load fluctuation characteristics of the power-added efficiency in the power amplifier circuit **101**. The vertical axis indicates the power-added efficiency, whose unit is "%", of the power amplifier circuit. The horizontal axis indicates the phase of an antenna **86***a* end, whose unit is "°" and which is obtained when VSWR (Voltage Standing Wave Ratio) is two.
[0102] As illustrated in FIGS. **20** to **22**, in the power amplifier circuit **101**, the signals RF**1** and RF**2**, which have a phase difference of approximately 90°, are amplified by the carrier amplifier **51** and the peak amplifier **52**, respectively. When the phase of the load **86** fluctuates, as illustrated in FIG. **20**, the output impedance Zc of the carrier amplifier **51** is shifted to the open OP side. At that time, as illustrated in FIG. **21**, the output impedance Zp of the peak amplifier **52** is easily shifted to the short SH side by the same shift amount.
[0103] Thus, as illustrated in FIG. **22**, compared with the fluctuation, which is illustrated by using curve Cn, of the power-added efficiency of an amplifier circuit including an amplifier of the related art such as a differential amplifier or a two-stage amplifier, the fluctuation, which is illustrated by using curve Cg, of the power-added efficiency of the power amplifier circuit **101** may be suppressed. That is, the resistance to load fluctuation in the power amplifier circuit **101** may be improved compared with that in an amplifier circuit of the related art.
[0104] In the present embodiment, the configuration, in which the 90° hybrid coupler **71** is formed by using the capacitors **203** and **204** and the inductors **213** and **214**, is described. However, the configuration is not limited to this. The 90° hybrid coupler **71** may have a configuration in which four quarter-wave lines are connected in a circle.
Second Embodiment
[0105] A power amplifier circuit **111** according to a second embodiment will be described. Points common to those in the first embodiment will not be described in the second embodiment and its subsequent embodiments, and only different points will be described. In particular, substantially the same operational effect caused by substantially the same configuration will not be described repeatedly.
[0106] FIG. **23** is a circuit diagram of a power amplifier circuit **111**. As illustrated in FIG. **23**, the power amplifier circuit **111** is different from the power amplifier circuit **101** according to the first embodiment in that an LC circuit **171** is disposed instead of the 90° hybrid coupler **71**.
[0107] The LC circuit **171** includes a capacitor **205** (third capacitor) and an inductor **215** (third inductor). The inductor **215** has a first end connected to the node N**1**, and a second end connected to the node N**3**. The capacitor **205** has a first end connected to the node N**2**, and a second end connected to the node N**3**.
[0108] A capacitor **303** has a first end connected to a power supply voltage VCC, and a second end connected to the ground. A line **313** has a first end connected to the power supply voltage VCC, and a second end connected to the output terminal **51***a* of the carrier amplifier **51**. A line **314** has a first end connected to the power supply voltage VCC, and a second end connected to the output terminal **52***a* of the peak amplifier **52**. The capacitor **303** and the lines **313** and **314** may be disposed in the power amplifier circuit **101**.
[0109] The output terminal **51***a* of the carrier amplifier **51** is grounded through the capacitor **301** and a line **311**. The output terminal **52***a* of the peak amplifier **52** is grounded through the capacitor

302 and a line **312**.

[0110] Thus, even the configuration, in which the LC circuit **171** which is simpler than the 90° hybrid coupler **71** is disposed, may achieve characteristics equivalent to those of the power amplifier circuit **101**. In other words, the power amplifier circuit **111** may also achieve power-added efficiency equivalent to that of the power amplifier circuit **101** in FIG. **1**.

Third Embodiment

[0111] A power amplifier circuit **102** according to a third embodiment will be described. FIG. **24** is a circuit diagram of a power amplifier circuit **102**. As illustrated in FIG. **24**, the power amplifier circuit **102** according to the third embodiment is different from the power amplifier circuit **101** in FIG. **1** according to the first embodiment in the connection form of the elements.

[0112] The capacitor **201** in the LC circuit **61** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1**. The inductor **211** has its first end connected to the node N**1**, and its second end connected to the ground.

[0113] The capacitor **202** in the LC circuit **62** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The inductor **212** has its first end connected to the node N**2**, and its second end connected to the ground.

[0114] The inductor **213** in the 90° hybrid coupler **71** has its first end, which is connected to the node N**2** and which is the input port Pin**2** supplied with the amplified signal RF**4**, and its second end, which is connected to the node N**3** and which is the output port Pout.

[0115] The capacitor **203** has its first end, which is connected to the node N**1** and which is the input port Pin**1** supplied with the amplified signal RF**3**, and its second end connected to the node N**3**.

[0116] The capacitor **204** has its first end connected to the node N**2**, and its second end which is the isolation port Piso. The isolation port Piso is not connected to circuit elements external to the 90° hybrid coupler **71**.

[0117] The inductor **214** has its first end connected to the node N**1**, and its second end connected to the second end of the capacitor **204**, and is electromagnetically coupled to the inductor **213**.

[0118] FIG. **25** is a diagram illustrating a simulation result of the frequency characteristics of the change in phase across the output terminals in the power amplifier circuit **102**. The vertical axis indicates the change in phase, whose unit is "°", across the output terminals. The horizontal axis indicates the frequency of the amplified signals RF**3** and RF**4**. The change in phase across the output terminals in this case is substantially the same as that in FIG. **4**.

[0119] In the power amplifier circuit **102**, the magnitude of the change in phase across the output terminals of the amplified signal RF**3** is greater than or equal to 45° and less than or equal to 135°. In the present embodiment, as illustrated in FIG. **25**, the magnitude of the change in phase across the output terminals of the amplified signal RF**3** is greater than or equal to 50° and less than or equal to 115° in the range from 1.68 GHz to 1.98 GHz. That is, the change in phase across the output terminals comparable to that in the power amplifier circuit **900** (see FIG. **3**) is achieved.

[0120] FIG. **26** is a diagram illustrating a simulation result of the frequency characteristics of the phase difference between the signals in the power amplifier circuit **102**. The vertical axis indicates the phase difference, whose unit is "°", between the signals. The horizontal axis indicates the frequency of the amplified signals RF**3** and RF**4**. The phase difference between the signals in this case is substantially the same as that in FIG. **6**.

[0121] In the power amplifier circuit **101**, the magnitude of the phase difference between the signals is greater than or equal to 60° and less than or equal to 120°. In the present embodiment, as illustrated in FIG. **26**, the magnitude of the phase difference between the signals is greater than or equal to 90° and less than or equal to 115° in the range from 1.68 GHz to 1.98 GHz. That is, the phase difference between the signals comparable to that in the power amplifier circuit **900** (see FIG. **5**) is achieved.

[0122] FIG. **27** is a circuit diagram of a power amplifier circuit **902** which is a reference example. As illustrated in FIG. **27**, compared with the power amplifier circuit **102** in FIG. **24**, the power

amplifier circuit **902** is a circuit in which the LC circuits **61** and **62** are not disposed.
[0123] FIGS. **28** and **29** are diagrams illustrating simulation results of the gain characteristics of the power amplifier circuits **102** and **902**, respectively. Reading of FIGS. **28** and **29** is substantially the same as that of FIGS. **8** to **10**.
[0124] The gain characteristics in FIGS. **28** and **29** are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The gain characteristics in the power amplifier circuit **102** (see FIG. **28**) are higher than that in the power amplifier circuit **902** (see FIG. **29**), and are comparable to that in the power amplifier circuit **900** (see FIG. **9**).
[0125] FIGS. **30** and **31** are diagrams illustrating simulation results of efficiency characteristics in the power amplifier circuits **102** and **902**, respectively. Reading of FIGS. **30** and **31** is substantially the same as that of FIGS. **11** to **13**.
[0126] The efficiency characteristics in FIGS. **30** and **31** are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The output power indicated by using a dotted line is the actual power in use.
[0127] The power-added efficiency of the power amplifier circuit **102** at the actual power in use is greater than that of the power amplifier circuit **902** at the actual power in use. The power-added efficiency of the power amplifier circuit **102** at the actual power in use is comparable to that of the power amplifier circuit **900** at the actual power in use.
[0128] FIGS. **32** and **33** are diagrams illustrating Smith charts showing simulation results of variations in impedance Zc relative to output power variations in the power amplifier circuits **102** and **902**, respectively.
[0129] The variations in impedance Zc relative to output power variations, which are illustrated in FIGS. **32** and **33**, are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The direction of an arrow indicated by using a dotted line indicates increase of output power.
[0130] In the power amplifier circuit **902** having only the 90° hybrid coupler **71**, impedance Zc varies along the imaginary number axis (see FIG. **33**).
[0131] In the power amplifier circuit **102** in which high-pass filter circuits of the LC circuits **61** and **62** are disposed, variations in impedance Zc relative to output power variations may be made close to those in the power amplifier circuit **900** (see FIGS. **15** and **32**).
[0132] FIGS. **34** and **35** are diagrams illustrating Smith charts showing simulation results of variations in impedance Zp relative to output power variations in the power amplifier circuits **102** and **902**, respectively.
[0133] The variations in impedance Zp relative to output power variations, which are illustrated in FIGS. **34** and **35**, are obtained in Doherty operation. The frequency of the signals RF**1** and RF**2** is 1.8 GHz. The direction of an arrow indicated by using a dotted line indicates increase of output power.
[0134] In the power amplifier circuit **902** having only the 90° hybrid coupler **71**, impedance Zp varies along the imaginary number axis (see FIG. **35**).
[0135] In the power amplifier circuit **102** in which the high-pass filter circuits of the LC circuits **61** and **62** are disposed, variations in impedance Zp relative to output power variations may be made close to that in the power amplifier circuit **900** (see FIGS. **18** and **34**).
[0136] In the present embodiment, the configuration, in which the 90° hybrid coupler **71** is formed by using the capacitors **203** and **204** and the inductors **213** and **214**, is described. However, the configuration is not limited to this. The 90° hybrid coupler **71** may have a configuration in which four quarter-wave lines are connected in a circle.
Fourth Embodiment
[0137] A power amplifier circuit **112** according to a fourth embodiment will be described. FIG. **36** is a circuit diagram of the power amplifier circuit **112**. As illustrated in FIG. **36**, the power amplifier circuit **112** is different from the power amplifier circuit **102** according to the third

embodiment in that the LC circuit **171** is disposed instead of the 90° hybrid coupler **71**.
[0138] The LC circuit **171** includes the capacitor **205** (third capacitor) and the inductor **215** (third inductor). The capacitor **205** has its first end connected to the node N**1**, and its second end connected to the node N**3**. The inductor **215** has its first end connected to the node N**2**, and its second end connected to the node N**3**.
[0139] The capacitor **303** and the lines **311** to **314** are substantially the same as the capacitor **303** and the lines **311** to **314**, respectively, in the power amplifier circuit **111** in FIG. **23**.
[0140] Thus, the configuration, having the LC circuit **171** which is simpler than the 90° hybrid coupler **71**, also achieves characteristics equivalent to those of the power amplifier circuit **102**.
Fifth Embodiment
[0141] A power amplifier circuit **122** according to a fifth embodiment will be described. FIG. **37** is a circuit diagram of the power amplifier circuit **122**. As illustrated in FIG. **37**, the power amplifier circuit **122** is different from the power amplifier circuit **102** according to the third embodiment in that the capacities of the capacitors in the 90° hybrid coupler **71** are variable.
[0142] Compared with the power amplifier circuit **102** in FIG. **24**, the power amplifier circuit **122** includes a 90° hybrid coupler **72** and an LC circuit **81** instead of the 90° hybrid coupler **71** and the capacitors **301** and **302**.
[0143] Compared with the 90° hybrid coupler **71** in FIG. **24**, the 90° hybrid coupler **72** includes capacitors **203***v* (third capacitor) and **204***v* (fourth capacitor) instead of the capacitors **203** and **204**, respectively. The capacity of the capacitor **203***v* and that of the capacitor **204***v* are variable.
[0144] The LC circuit **81** includes an inductor **82** and a capacitor **83**, and is disposed between the node N**3** and the load **86** (not illustrated). The inductor **82** has a first end connected to the output port Pout of the 90° hybrid coupler **72**, and a second end connected to the load **86**. The capacitor **83** has a first end connected to the second end of the inductor **82**, and a second end connected to the ground.
[0145] Adjustment of the capacity of the capacitor **203***v* and that of the capacitor **204***v* enables the frequency characteristics of the change in phase across the output terminals in the power amplifier circuit **122** to be changed. Specifically, the frequency characteristics of the change in phase across the output terminals in FIG. **25** may be shifted in the vertical-axis direction.
[0146] The capacity of the capacitor **203***v* and that of the capacitor **204***v* are adjusted appropriately in accordance with the frequency band of the signals RF**1** and RF**2**. Thus, optimal change in phase across the output terminals is achieved in the frequency band. This enables the power amplifier circuit **122** to operate in a wide band excellently. In the present embodiment, the configuration of the power amplifier circuit **122**, in which the capacities of the capacitors **203** and **204** of the 90° hybrid coupler **71** in the power amplifier circuit **102** are made variable, is described. However, the configuration is not limited to this. The capacities of the capacitors **203** and **204** of the 90° hybrid coupler **71** in the power amplifier circuit **101** may be made variable.
Sixth Embodiment
[0147] A power amplifier circuit **132** according to a sixth embodiment will be described. FIG. **38** is a circuit diagram of the power amplifier circuit **132**. As illustrated in FIG. **38**, the power amplifier circuit **132** is different from the power amplifier circuit **102** according to the third embodiment in that the isolation port Piso in the 90° hybrid coupler **71** may be grounded through a resistive element.
[0148] Compared with the power amplifier circuit **102** in FIG. **24**, the power amplifier circuit **132** includes the LC circuit **81**, a switch **401**, and a resistive element **402** instead of the capacitors **301** and **302**.
[0149] In the power amplifier circuit **132**, the resistive element **402** and the switch **401** are connected in series between the isolation port Piso of the 90° hybrid coupler **71** and the ground.
[0150] In the present embodiment, the switch **401** has a first end connected to the isolation port Piso of the 90° hybrid coupler **71**, and a second end connected to the ground through the resistive

element **402**. The switch **401** may be connected, at its first end, to the isolation port Piso through the resistive element **402**, and may be connected, at its second end, to the ground.
[0151] The LC circuit **81** includes the inductor **82** and the capacitor **83**, and is disposed between the node N**3** and the antenna **86***a* (load). The inductor **82** has its first end connected to the output port Pout of the 90° hybrid coupler **71**, and its second end connected to the antenna **86***a*. The capacitor **83** has its first end connected to the second end of the inductor **82**, and its second end connected to the ground.
[0152] When the switch **401** is ON and the isolation port Piso is grounded through the resistive element **402**, the power amplifier circuit **132** operates in a first mode (hereinafter may be referred to as balance mode).
[0153] When the switch **401** is OFF and the isolation port Piso is electrically detached from the ground, the power amplifier circuit **132** operates in a second mode (hereinafter may be referred to as Doherty mode).
[0154] FIG. **39** is a diagram illustrating exemplary load fluctuation characteristics of the power-added efficiency in the power amplifier circuit **132**. Reading of FIG. **39** is substantially the same as that of FIG. **22**.
[0155] Curves Cd and Cb indicate variations of the power-added efficiency which are obtained when the phase of the antenna **86***a* end is varied in Doherty mode and balance mode, respectively.
[0156] Curve Cr indicates variations of the power-added efficiency which are obtained when the phase of the antenna **86***a* end is varied in an amplifier circuit including an amplifier of the related art, such as a differential amplifier or a two-stage amplifier (hereinafter may be referred to as an amplifier circuit of the related art).
[0157] Curve Cr shows power-added efficiency close to 50 at or near a phase of 90°, but shows power-added efficiency of about 30 at or near a phase of 270°. Therefore, the amplifier circuit of the related art has a large amount of change of the power-added efficiency when the phase of the antenna **86***a* end fluctuates.
[0158] Curve Cb shows power-added efficiency of 35 or greater when the phase is varied by 360°, and shows that the range of fluctuation of the power-added efficiency is within about 3. Therefore, balance mode is a mode suitable for a specification which requires that fluctuation in the characteristics for the load impedance at the antenna **86***a* end be small.
[0159] Compared with curve Cb, curve Cd shows a slightly larger range of fluctuation of the power-added efficiency, but shows higher power-added efficiency at all the phases. Curve Cd shows a smaller range of fluctuation of the power-added efficiency than curve Cr. Therefore, Doherty mode is a mode suitable for a specification which requires excellent power-added efficiency while maintaining a constant range of fluctuation in the characteristics for the load impedance at the antenna **86***a* end.
[0160] In the present embodiment, the configuration of the power amplifier circuit **132**, in which, in the power amplifier circuit **102**, the resistive element **402** and the switch **401** are connected in series between the isolation port Piso of the 90° hybrid coupler **71** and the ground, is described. However, the configuration is not limited to this. A configuration, in which, in the power amplifier circuit **101**, the resistive element **402** and the switch **401** are connected in series between the isolation port Piso of the 90° hybrid coupler **71** and the ground, may be employed.
Seventh Embodiment
[0161] A power amplifier circuit **141** according to a seventh embodiment will be described. FIG. **40** is a circuit diagram of the power amplifier circuit **141**. The power amplifier circuit **111** in FIG. **23** may have a difficulty in control of the load impedance of the carrier amplifier **51** and the peak amplifier **52** in the frequency band of the second harmonic in inverse class-F operation. Therefore, harmonic distortion characteristics may degrade.
[0162] In the present embodiment, the power amplifier circuit **141** is different from the power amplifier circuit **111** in FIG. **23** in the following points: the circuit configuration of the LC circuit

62; a point in which the inductor **215** is not disposed; and a point in which an LC circuit **181** is further included.

[0163] Compared with the LC circuit **62** in the power amplifier circuit **111**, the LC circuit **62** in the power amplifier circuit **141** includes a capacitor **202***v* (second capacitor) instead of the capacitor **202**, and further includes an inductor **225** (fifth inductor).

[0164] The inductor **212** in the power amplifier circuit **141** has its first end connected to the output terminal **52***a* of the peak amplifier **52**, and its second end connected to the node N**2**.

[0165] The inductor **225** and the capacitor **202***v* are connected in series between the node N**2** and the ground.

[0166] In the present embodiment, the inductor **225** has a first end connected to the node N**2**, and a second end connected to the ground through the capacitor **202***v*. The inductor **225** may be connected, at its first end, to the node N**2** through the capacitor **202***v*, and may be grounded at its second end.

[0167] FIG. **41** is a diagram illustrating an exemplary capacitor **202***v*. As illustrated in FIG. **41**, the electrostatic capacity of the capacitor **202***v* is variable. In the present embodiment, the capacitor **202***v* includes capacitors **2021** and **2022** and a switch **2023**. For example, the electrostatic capacity of the capacitor **2021** is larger than that of the capacitor **2022**.

[0168] The capacitor **2021** has a first end connected to the second end of the inductor **225**, and a second end connected to the ground. The capacitor **2022** has a first end connected to the second end of the inductor **225**, and a second end connected to the ground through the switch **2023**. The capacitor **2022** may be connected, at its first end, to the second end of the inductor **225** through the switch **2023**, and may be grounded at its second end.

[0169] The power amplifier circuit **141** may be used, for example, in the high frequency mode and the low frequency mode. The high frequency mode and the low frequency mode are suitable for amplification of input signals in a first frequency band and input signals in a second frequency band, respectively. The second frequency band is lower than the first frequency band. The first frequency band and the second frequency band are, for example, bands of approximately 0.7 to 0.9 GHz.

[0170] The switch **2023** is OFF in the high frequency mode, and is ON in the low frequency mode. That is, the electrostatic capacity of the capacitor **202***v* in the low frequency mode is larger than that in the high frequency mode.

[0171] As illustrated in FIG. **40**, the capacitor **205** (third capacitor) has its first end connected to the node N**2**, and its second end connected to the node N**1** through the node N**3**.

[0172] The LC circuit **181** is connected between the node N**3** and the load **86**. In detail, the LC circuit **181** includes a tank circuit **182**, a capacitor **185***v*, and a capacitor **186**. The tank circuit **182** includes a capacitor **183***v* (fifth capacitor) and an inductor **184** (sixth inductor).

[0173] The capacitor **186** in the LC circuit **181** has a first end connected to the node N**3**, and a second end. The capacitor **186** cuts DC components of an amplified signal obtained by combining the amplified signals RF**3** and RF**4** with each other.

[0174] The inductor **184** in the tank circuit **182** has a first end connected to the second end of the capacitor **186**, and a second end connected to the ground through the load **86**.

[0175] The capacitor **183***v* has a first end connected to the second end of the capacitor **186**, and a second end connected to the second end of the inductor **184**.

[0176] FIG. **42** is a diagram illustrating an exemplary capacitor **183***v*. As illustrated in FIG. **42**, the electrostatic capacity of the capacitor **183***v* is variable. In the present embodiment, the capacitor **183***v* includes capacitors **1831** and **1832** and a switch **1833**. For example, the electrostatic capacity of the capacitor **1831** is larger than that of the capacitor **1832**.

[0177] The capacitor **1831** has a first end connected to the second end of the capacitor **186**, and a second end connected to the second end of the inductor **184**. The capacitor **1832** has a first end connected to the second end of the capacitor **186**, and a second end connected to the second end of

the inductor **184** through the switch **1833**. The capacitor **1832** may be connected, at its first end, to the second end of the capacitor **186** through the switch **1833**, and may be connected, at its second end, to the second end of the inductor **184**.
[0178] The switch **1833** is OFF in the high frequency mode and is ON in the low frequency mode. That is, the electrostatic capacity of the capacitor **183***v* in the low frequency mode is larger than that in the high frequency mode.
[0179] As illustrated in FIG. **40**, the capacitor **185***v* has a first end connected to the second end of the inductor **184**, and a second end connected to the ground. The capacitor **185***v* converts the impedance.
[0180] The electrostatic capacity of the capacitor **185***v* is variable. In the present embodiment, the capacitor **185***v* has a configuration substantially the same as that of the capacitor **202***v*. The electrostatic capacity of the capacitor **185***v* in the low frequency mode is larger than that in the high frequency mode.
[0181] FIGS. **43** and **44** are diagrams illustrating Smith charts showing simulation results of variations in impedances Zp and Zc, respectively, relative to frequency variations in the power amplifier circuit **141**.
[0182] FIG. **43** illustrates curves FHZp, FLZp, HHZp, and HLZp. Curves FHZp and FLZp show variations in impedance Zp for the fundamental in the first frequency band and the second frequency band, respectively. Curves HHZp and HLZp show variations in impedance Zp for the second harmonic based on the fundamental in the first frequency band and the second frequency band, respectively.
[0183] FIG. **44** illustrates curves FHZc, FLZc, HHZc, and HLZc. Curves FHZc and FLZc show variations in impedance Zc for the fundamental in the first frequency band and the second frequency band, respectively. Curves HHZc and HLZc show variations in impedance Zc for the second harmonic based on the fundamental in the first frequency band and the second frequency band, respectively.
[0184] FIG. **45** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance ZP**2** relative to frequency variations in the power amplifier circuit **141**. FIG. **45** illustrates curves FHZP**2**, FLZP**2**, HHZP**2**, and HLZP**2**. Curves FHZP**2** and FLZP**2** show variations in impedance ZP**2** for the fundamental in the first frequency band and the second frequency band, respectively. Curves HHZP**2** and HLZP**2** show variations in impedance ZP**2** for the second harmonic based on the fundamental in the first frequency band and the second frequency bands, respectively. Impedance ZP**2** is impedance as seen from the capacitor **186** to the carrier amplifier **51** and the peak amplifier **52**.
[0185] FIG. **46** is a diagram illustrating a Smith chart showing a simulation result of variations in impedance ZP**1** relative to frequency variations in the power amplifier circuit **141**. FIG. **46** illustrates curves FHZP**1**, FLZP**1**, HHZP**1**, and HLZP**1**. Curves FHZP**1** and FLZP**1** show variations in impedance ZP**1** for the fundamental in the first frequency band and the second frequency band, respectively. Curves HHZP**1** and HLZP**1** show variations in impedance ZP**1** for the second harmonic based on the fundamental in the first frequency band and the second frequency band, respectively. Impedance ZP**1** is impedance as seen from the node N**3** to the load **86**.
[0186] As illustrated in FIGS. **43** and **44**, curves FHZp, FLZp, FHZc, and FLZc are positioned near 1.0+0.0j. That is, control may be exerted so that impedance Zp and impedance Zc for the fundamental are comparable to the characteristic impedance.
[0187] Curves HHZp, HLZp, HHZc, and HLZc are positioned near the open OP. That is, control may be exerted so that the magnitudes of impedance Zp and impedance Zc for the second harmonic increase.
[0188] As illustrated in FIG. **45**, the configuration, in which the LC circuit **62** further including the inductor **225** is disposed at the output of the peak amplifier **52** (see FIG. **40**), causes curves HHZP**2**

and HLZP**2** to be positioned near the short SH. That is, the magnitude of impedance ZP**2** for the second harmonic is small.

[0189] In contrast, as illustrated in FIG. **46**, the configuration, in which the tank circuit **182** is disposed between the node N**3** and the load **86** (see FIG. **40**), enables control to be exerted so that curves HHZP**1** and HLZP**1** are positioned near the open OP.

[0190] Thus, the difference between the phase of impedance ZP**2** for the second harmonic and that of impedance ZP**1** for the second harmonic may be set to approximately 180°. That is, for the second harmonic, the difference between the phase of traveling waves, which are transmitted from the carrier amplifier **51** and the peak amplifier **52** to the load **86**, and the phase of reflected waves, which are obtained through reflection of the traveling waves and are transmitted to the carrier amplifier **51** and the peak amplifier **52**, may be set to approximately 180°. This causes the traveling waves and the reflected waves of the second harmonic to cancel each other, achieving suppression of degradation of the harmonic distortion characteristics.

[0191] FIG. **47** is a diagram illustrating exemplary variations in loss relative to frequency variations in the LC circuit **181**. The vertical axis indicates loss whose unit is "dBm". The horizontal axis indicates frequency whose unit is "GHz".

[0192] FIG. **47** illustrates curves LLs and HLs. Curve LLS shows variations in loss relative to frequency variations when a signal passes through the LC circuit **181** in the case where the capacitor **185***v* has an electrostatic capacity of the low frequency mode.

[0193] Curve HLs shows variations in loss relative to frequency variations when a signal passes through the LC circuit **181** in the case where the capacitor **185***v* has an electrostatic capacity of the high frequency mode.

[0194] As illustrated in FIG. **47**, in the fundamental band FB, mainly by switching the capacitor **185***v*, occurrence of loss caused when fundamental waves pass through the LC circuit **181** may be suppressed both in the high frequency mode and in the low frequency mode.

[0195] In the second harmonic band HB, mainly by switching the capacitor **183***v*, the second harmonic waves based on the fundamental waves may be attenuated excellently both in the high frequency mode and in the low frequency mode.

[0196] The configuration, in which the capacitors **183***v* and **185***v* enable switching between the low frequency mode and the high frequency mode, achieves excellent attenuation of the second harmonic based on the fundamental while degradation of the bandpass characteristics of the fundamental is suppressed in a broad frequency band.

[0197] In the present embodiment, the configuration, in which the electrostatic capacities of the capacitors **183***v*, **185***v*, and **202***v* are variable, is described. However, the configuration is not limited to this. At least one of the electrostatic capacities of the capacitors **183***v*, **185***v*, and **202***v* may be fixed.

[0198] The exemplary embodiments of the present disclosure are described above. In the power amplifier circuits **101** and **111**, the carrier amplifier **51** amplifies the signal RF**1**, and outputs the amplified signal RF**3** from the output terminal **51***a*. The peak amplifier **52** amplifies the signal RF**2** different in phase from the signal RF**1**, and outputs the amplified signal RF**4** from the output terminal **52***a*. The inductor **211** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1**. The capacitor **201** has its first end connected to the node N**1**, and its second end connected to the ground. The inductor **212** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The capacitor **202** has its first end connected to the node N**2**, and its second end connected to the ground. The inductors **213** and **215** each has its first end connected to the node N**1**, and its second end connected to the node N**3** connected to the load **86**. The capacitors **203** and **205** each have its first end connected to the node N**2**, and its second end connected to the node N**3**.

[0199] A configuration, in which the inductors **211** and **212** and the capacitors **201** and **202** are not disposed, has a difficulty in adjustment of the phase difference between the signals and the change in phase across the output terminals. As described above, the configuration in which the inductor **211** series-connected between the output terminal **51***a* and the node N**1**, the capacitor **201** shunt-connected to the node N**1**, the inductor **212** series-connected between the output terminal **52***a* and the node N**2**, and the capacitor **202** shunt-connected to the node N**2** are disposed, achieves adequate adjustment of the phase difference between the signals and the change in phase across the output terminals. Thus, operation close to that of a Doherty amplifier circuit may be implemented, achieving enhancement of the power-added efficiency and suppression of variations of the power-added efficiency relative to fluctuations in the load impedance. In addition, since no quarter-wave lines are used, the circuit sizes of the power amplifier circuits **101** and **111** may be decreased. Therefore, a power amplifier circuit, which enables a reduction in size, high efficiency, and high resistance to fluctuations in the load impedance to be achieved, may be provided.

[0200] In the power amplifier circuit **101**, the carrier amplifier **51** amplifies the signal RF**1**, and outputs the amplified signal RF**3** from the output terminal **51***a*. The peak amplifier **52** amplifies the signal RF**2** different in phase from the signal RF**1**, and outputs the amplified signal RF**4** from the output terminal **52***a*. The inductor **211** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1**. The capacitor **201** has its first end connected to the node N**1**, and its second end connected to the ground. The inductor **212** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The capacitor **202** has its first end connected to the node N**2**, and its second end connected to the ground. The 90° hybrid coupler **71** has the input port Pin**1**, which is connected to the node N**1** and which is supplied with the amplified signal RF**3**, the input port Pin**2**, which is connected to the node N**2** and which is supplied with the amplified signal RF**4**, the output port Pout, which is connected to the node N**3** connected to the load **86** and from which the output signal RFout is outputted, and the isolation port Piso.

[0201] A configuration, in which the inductors **211** and **212** and the capacitors **201** and **202** are not disposed, has a difficulty in adjustment of the phase difference between the signals and the change in phase across the output terminals. As described above, the configuration, in which the inductor **211** series-connected between the output terminal **51***a* and the node N**1**, the capacitor **201** shunt-connected to the node N**1**, the inductor **212** series-connected between the output terminal **52***a* and the node N**2**, and the capacitor **202** shunt-connected to the node N**2** are disposed, achieves adequate adjustment of the phase difference between the signals and the change in phase across the output terminals. Thus, operation close to that of a Doherty amplifier circuit may be implemented, achieving enhancement of the power-added efficiency and suppression of variations of the power-added efficiency relative to fluctuations in the load impedance. In addition, since no quarter-wave lines are used, the circuit size of the power amplifier circuit **101** may be decreased. Therefore, a power amplifier circuit, which enables a reduction in size, high efficiency, and high resistance to fluctuations in the load impedance to be achieved, may be provided. In addition, since the isolation port Piso is not connected to any terminator, output matching loss caused by a terminator may be suppressed. Use of the 90° hybrid coupler **71** achieves enhancement of accuracy of adjustment of the phase difference between the signals and the change in phase across the output terminals.

[0202] In the power amplifier circuits **102** and **112**, the carrier amplifier **51** amplifies the signal RF**1**, and outputs the amplified signal RF**3** from the output terminal **51***a*. The peak amplifier **52** amplifies the signal RF**2** different in phase from the signal RF**1**, and outputs the amplified signal RF**4** from the output terminal **52***a*. The capacitor **201** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1**. The inductor **211** has its first end connected to the node N**1**, and its second end connected to the ground. The capacitor **202** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The inductor **212** has its first end connected to the node N**2**, and its second end connected to the ground. The capacitors **203** and **205** each have its first end connected to the node N**1**, and its second end connected to the node N**3** connected to the load **86**. The inductors **213** and **215** each has its first end connected to the node N**2**, and its second end connected to the node N**3**.

[0203] A configuration, in which the inductors **211** and **212** and the capacitors **201** and **202** are not disposed, has a difficulty in adjustment of the phase difference between the signals and the change in phase across the output terminals. As described above, the configuration, in which the capacitor **201** series-connected between the output terminal **51***a* and the node N**1**, the inductor **211** shunt-connected to the node N**1**, the capacitor **202** series-connected between the output terminal **52***a* and the node N**2**, and the inductor **212** shunt-connected to the node N**2** are disposed, achieves adequate adjustment of the phase difference between the signals and the change in phase across the output terminals. Thus, operation close to that of a Doherty amplifier circuit may be implemented, achieving enhancement of the power-added efficiency and suppression of variations of the power-added efficiency relative to fluctuations in the load impedance. In addition, since no quarter-wave lines are used, the circuit sizes of the power amplifier circuits **102** and **112** may be decreased. Therefore, a power amplifier circuit, which enables a reduction in size, high efficiency, and high resistance to fluctuations in the load impedance to be achieved, may be provided.

[0204] In the power amplifier circuit **102**, the carrier amplifier **51** amplifies the signal RF**1**, and outputs the amplified signal RF**3** from the output terminal **51***a*. The peak amplifier **52** amplifies the signal RF**2** different in phase from the signal RF**1**, and outputs the amplified signal RF**4** from the output terminal **52***a*. The capacitor **201** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1**. The inductor **211** has its first end connected to the node N**1**, and its second end connected to the ground. The capacitor **202** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The inductor **212** has its first end connected to the node N**2**, and its second end connected to the ground. The 90° hybrid coupler **71** has the input port Pin**1**, which is connected to the node N**1** and which is supplied with the amplified signal RF**3**, the input port Pin**2**, which is connected to the node N**2** and which is supplied with the amplified signal RF**4**, the output port Pout, which is connected to the node N**3** connected to the ground through the load **86** and from which the output signal RFout is outputted, and the isolation port Piso.

[0205] A configuration, in which the inductors **211** and **212** and the capacitors **201** and **202** are not disposed, has a difficulty in adjustment of the phase difference between the signals and the change in phase across the output terminals. As described above, the configuration, in which the capacitor **201** series-connected between the output terminal **51***a* and the node N**1**, the inductor **211** shunt-connected to the node N**1**, the capacitor **202** series-connected between the output terminal **52***a* and the node N**2**, and the inductor **212** shunt-connected to the node N**2** are disposed, achieves adequate adjustment of the phase difference between the signals and the change in phase across the output terminals. Thus, operation close to that of a Doherty amplifier circuit may be implemented, achieving enhancement of the power-added efficiency and suppression of variations of the power-added efficiency relative to fluctuations in the load impedance. In addition, since no quarter-wave lines are used, the circuit size of the power amplifier circuit **102** may be decreased. Therefore, a power amplifier circuit, which enables a reduction in size, high efficiency, and high resistance to fluctuations in the load impedance to be achieved, may be provided. In addition, since the isolation port Piso is not connected to any terminator, output matching loss caused by a terminator may be suppressed. Use of the 90° hybrid coupler **71** achieves enhancement of accuracy of adjustment of the phase difference between the signals and the change in phase across the output terminals.

[0206] In the power amplifier circuit **101**, the capacitor **204** has its first end connected to the node N**1**, and its second end which is the isolation port Piso. The inductor **214** has its first end connected to the node N**2**, and its second end connected to the second end of the capacitor **204**. The inductor **213** is electromagnetically coupled to the inductor **214**.

[0207] Such a configuration enables the 90° hybrid coupler **71** to be implemented by using the capacitors **203** and **204** and the inductors **213** and **214** without necessarily use of a quarter-wave line, achieving a reduction in circuit size of the power amplifier circuit **101**. Implementation of the 90° hybrid coupler **71** achieves enhancement of accuracy of adjustment of the phase difference between the signals and the change in phase across the output terminals.

[0208] In the power amplifier circuit **102**, the capacitor **204** has its first end connected to the node N**2**, and its second end which is the isolation port Piso. The inductor **214** has its first end connected to the node N**1**, and its second end connected to the second end of the capacitor **204**. The inductor **213** is electromagnetically coupled to the inductor **214**.

[0209] Such a configuration enables the 90° hybrid coupler **71** to be implemented by using the capacitors **203** and **204** and the inductors **213** and **214** without use of a quarter-wave line, achieving a reduction in circuit size of the power amplifier circuit **102**. Implementation of the 90° hybrid coupler **71** achieves enhancement of accuracy of adjustment of the phase difference between the signals and the change in phase across the output terminals.

[0210] In the power amplifier circuits **101**, **102**, **111**, and **112**, the magnitude of the difference between the change in phase, which is obtained when the amplified signal RF**3** outputted from the output terminal **51***a* passes through the node N**1**, the node N**3**, and the load **86**, and the change in phase, which is obtained when the amplified signal RF**4** outputted from the output terminal **52***a* passes through the node N**2**, the node N**3**, and the load **86**, is greater than or equal to 60° and less than or equal to 120°.

[0211] With a circuit size smaller than that of the power amplifier circuit **900** which is a Doherty amplifier circuit of the related art and which has the quarter-wave line **42**, such a configuration enables the phase difference between the signals in the power amplifier circuits **101**, **102**, **111**, and **112** to be equivalent to that of the power amplifier circuit **900**.

[0212] In the power amplifier circuits **101**, **102**, **111**, and **112**, the magnitude of the change in phase of the amplified signal RF**3** is greater than or equal to 45° and less than or equal to 135° when the amplified signal RF**3** passes from the output terminal **51***a* through the nodes N**1** and N**2** to the output terminal **52***a*.

[0213] With a circuit size smaller than that of the power amplifier circuit **900** which is a Doherty amplifier circuit of the related art and which has the quarter-wave line **42**, such a configuration enables the change in phase across the output terminals in the power amplifier circuits **101**, **102**, **111**, and **112** to be equivalent to that of the power amplifier circuit **900**.

[0214] In the power amplifier circuit **122**, each of the capacities of the capacitor **203***v* and the capacitor **204***v* is variable.

[0215] Thus, adjustment of the capacities of the capacitor **203***v* and the capacitor **204***v* enables the frequency characteristics of the change in phase across the output terminals in the power amplifier circuit **122** to be changed. Specifically, appropriate adjustment of the capacities of the capacitor **203***v* and the capacitor **204***v* in accordance with the frequency band of the signals RF**1** and RF**2** enables optimal change in phase across the output terminals to be achieved in the frequency band. Thus, the power amplifier circuit **122** may operate excellently in a wide band.

[0216] In the power amplifier circuit **132**, the switch **401** and the resistive element **402** are connected in series between the isolation port Piso and the ground.

[0217] Such a configuration enables operation as a balanced amplifier to be performed by switching on the switch **401**. Thus, the power amplifier circuit **132** may operate in the balance mode suitable for a specification which requires that fluctuations in the characteristics for the impedance of the load at the antenna **86***a* end be small. Switching off the switch **401** enables operation as an amplifier circuit equivalent to the power amplifier circuit **102** to be performed. Thus, the power amplifier circuit **132** may operate in the Doherty mode suitable for a specification which requires excellent power-added efficiency.

[0218] In the power amplifier circuit **141**, the carrier amplifier **51** amplifies the signal RF**1**, and outputs the amplified signal RF**3** from the output terminal **51***a*. The peak amplifier **52** amplifies the signal RF**2** different in phase from the signal RF**1**, and outputs the amplified signal RF**4** from the output terminal **52***a*. The inductor **211** has its first end connected to the output terminal **51***a*, and its second end connected to the node N**1** connected to the load **86**. The capacitor **201** has its first end

connected to the node N**1**, and its second end connected to the ground. The inductor **212** has its first end connected to the output terminal **52***a*, and its second end connected to the node N**2**. The capacitor **205** has its first end connected to the node N**2**, and its second end connected to the node N**1**. The inductor **225** and the capacitor **202***v* are connected in series between the node N**2** and the ground.
[0219] Thus, the configuration, in which the inductor **225** and the capacitor **202***v* are connected in series between the node N**2** and the ground, enables impedance Zp and impedance Zc for the second harmonic based on the fundamental to be made close to the open OP. Thus, the power of the second harmonic which is outputted to the load **86** side may be reduced, achieving suppression of degradation of the harmonic distortion characteristics. A configuration, in which the inductors **211**, **212**, and **225** and the capacitors **201** and **202***v* are not disposed, has a difficulty in adjustment of the phase difference between the signals and the change in phase across the output terminals. As described above, the configuration, in which the inductor **211** series-connected between the output terminal **51***a* and the node N**1**, the capacitor **201** shunt-connected to the node N**1**, the inductor **212** series-connected between the output terminal **52***a* and the node N**2**, and the inductor **225** and the capacitor **202***v* connected in series between the node N**2** and the ground are disposed, achieves adequate adjustment of the phase difference between the signals and the change in phase across the output terminals. Thus, operation close to that of a Doherty amplifier circuit may be implemented, achieving enhancement of the power-added efficiency and suppression of variations of the power-added efficiency relative to fluctuations in the load impedance. In addition, since no quarter-wave lines are used, the circuit size of the power amplifier circuit **141** may be decreased. Further, the configuration, in which the inductor **215** in the power amplifier circuit **111** (see FIG. **23**) is not disposed in the power amplifier circuit **141**, achieves a further smaller circuit size of the power amplifier circuit **141**. Therefore, a power amplifier circuit, which enables a reduction in size, high efficiency, and high resistance to fluctuations in the load impedance to be achieved, may be provided.
[0220] In the power amplifier circuit **141**, the inductor **184** has its first end connected to the node N**1**, and its second end connected to the load **86**. The capacitor **183***v* has its first end connected to the first end of the inductor **184**, and its second end connected to the second end of the inductor **184**.
[0221] Impedance ZP**2** as seen from the capacitor **186** to the carrier amplifier **51** and the peak amplifier **52** is positioned near the short SH because of arrangement of the LC circuit **62** including the inductor **225**. The configuration described above enables impedance ZP**1** as seen from the node N**3** to the load **86** to be made close to the open OP, and enables the difference between the phase of impedance ZP**2** for the second harmonic and that of impedance ZP**1** for the second harmonic to be set to approximately 180°. Thus, degradation of the harmonic distortion characteristics may be suppressed.
[0222] In the power amplifier circuit **141**, the electrostatic capacity of the capacitor **202***v* is variable.
[0223] Thus, the configuration, in which the resonant frequency of the inductor **225** and the capacitor **202***v* may be varied through variations of the electrostatic capacity of the capacitor **202***v*, achieves a wider frequency band in which the second harmonic based on the fundamental is attenuated excellently while degradation of the bandpass characteristics of the fundamental is suppressed.
[0224] In the power amplifier circuit **141**, the electrostatic capacity of the capacitor **183***v* is variable.
[0225] Thus, the configuration, in which the resonant frequency of the inductor **184** and the capacitor **183***v* may be varied through variations of the electrostatic capacity of the capacitor **183***v*, achieves a wider frequency band in which the second harmonic based on the fundamental is attenuated excellently while degradation of the bandpass characteristics of the fundamental is

suppressed.

[0226] The embodiments are described above to facilitate understanding of the present disclosure, not to limit the interpretation of the present disclosure. The present disclosure may be changed/improved without departing from its gist, and encompasses its equivalents. That is, embodiments obtained by those skilled in the art appropriately changing the design of the embodiments are encompassed in the scope of the present disclosure as long as they have features of the present disclosure. For example, the components included in the embodiments, and their arrangement, materials, conditions, shapes, sizes, and the like are not limited to the illustrated ones, and may be changed appropriately. Needless to say, the embodiments are exemplary, and partial replacement or combination of configurations in different embodiments may be made. These are encompassed in the scope of the present disclosure as long as these have features of the present disclosure.

<1>

[0227] A power amplifier circuit comprising: [0228] a carrier amplifier that amplifies a first signal, and that outputs a first amplified signal from a first output terminal; [0229] a peak amplifier that amplifies a second signal different in phase from the first signal, and that outputs a second amplified signal from a second output terminal; [0230] a first inductor that has a first end connected to the first output terminal, and a second end connected to a first node; [0231] a first capacitor that has a first end connected to the first node, and a second end connected to a ground; [0232] a second inductor that has a first end connected to the second output terminal, and a second end connected to a second node; [0233] a second capacitor that has a first end connected to the second node, and a second end connected to the ground; [0234] a third inductor that has a first end connected to the first node, and a second end connected to a third node connected to a load; and [0235] a third capacitor that has a first end connected to the second node, and a second end connected to the third node.

<2>

[0236] A power amplifier circuit comprising: [0237] a carrier amplifier that amplifies a first signal, and that outputs a first amplified signal from a first output terminal; [0238] a peak amplifier that amplifies a second signal different in phase from the first signal, and that outputs a second amplified signal from a second output terminal; [0239] a first inductor that has a first end connected to the first output terminal, and a second end connected to a first node; [0240] a first capacitor that has a first end connected to the first node, and a second end connected to a ground; [0241] a second inductor that has a first end connected to the second output terminal, and a second end connected to a second node; [0242] a second capacitor that has a first end connected to the second node, and a second end connected to the ground; and [0243] a 90° hybrid coupler that has a first input port, a second input port, an output port, and an isolation port, the first input port being connected to the first node and being supplied with the first amplified signal, the second input port being connected to the second node and being supplied with the second amplified signal, the output port being connected to a third node connected to a load and being a port from which an output signal is outputted.

<3>

[0244] A power amplifier circuit comprising: [0245] a carrier amplifier that amplifies a first signal, and that outputs a first amplified signal from a first output terminal; [0246] a peak amplifier that amplifies a second signal different in phase from the first signal, and that outputs a second amplified signal from a second output terminal; [0247] a first capacitor that has a first end connected to the first output terminal, and a second end connected to a first node; [0248] a first inductor that has a first end connected to the first node, and a second end connected to a ground; [0249] a second capacitor that has a first end connected to the second output terminal, and a second end connected to a second node; [0250] a second inductor that has a first end connected to the second node, and a second end connected to the ground; [0251] a third capacitor that has a first end connected to the first node, and a second end connected to a third node connected to a load; and [0252] a third inductor that has a first end connected to the second node, and a second end connected to the third node.

<4>

[0253] A power amplifier circuit comprising: [0254] a carrier amplifier that amplifies a first signal, and that outputs a first amplified signal from a first output terminal; [0255] a peak amplifier that amplifies a second signal different in phase from the first signal, and that outputs a second amplified signal from a second output terminal; [0256] a first capacitor that has a first end connected to the first output terminal, and a second end connected to a first node; [0257] a first inductor that has a first end connected to the first node, and a second end connected to a ground; [0258] a second capacitor that has a first end connected to the second output terminal, and a second end connected to a second node; [0259] a second inductor that has a first end connected to the second node, and a second end connected to the ground; and [0260] a 90° hybrid coupler that has a first input port, a second input port, an output port, and an isolation port, the first input port being connected to the first node and being supplied with the first amplified signal, the second input port being connected to the second node and being supplied with the second amplified signal, the output port being connected to a third node connected to a load and being a port from which an output signal is outputted.

<5>

[0261] The power amplifier circuit according to <1>, further comprising: [0262] a fourth capacitor that has a first end connected to the first node, and a second end which is an isolation port; and [0263] a fourth inductor that has a first end connected to the second node, and a second end connected to the second end of the fourth capacitor, [0264] wherein the third inductor is electromagnetically coupled to the fourth inductor.

<6>

[0265] The power amplifier circuit according to <3>, further comprising: [0266] a fourth capacitor that has a first end connected to the second node, and a second end which is an isolation port; and [0267] a fourth inductor that has a first end connected to the first node, and a second end connected to the second end of the fourth capacitor, [0268] wherein the third inductor is electromagnetically coupled to the fourth inductor.

<7>

[0269] The power amplifier circuit according to any one of <1> to <6>, [0270] wherein a magnitude of a difference between a change in phase of the first amplified signal and a change in phase of the second amplified signal is greater than or equal to 60° and less than or equal to 120°, the change in phase of the first amplified signal being obtained when the first amplified signal outputted from the first output terminal passes through the first node, the third node, and the load, the change in phase of the second amplified signal being obtained when the second amplified signal outputted from the second output terminal passes through the second node, the third node, and the load.

<8>

[0271] The power amplifier circuit according to any one of <1> to <7>, [0272] wherein a magnitude of a change in phase of the first amplified signal is greater than or equal to 45° and less than or equal to 135° when the first amplified signal passes from the first output terminal through the first node and the second node to the second output terminal.

<9>

[0273] The power amplifier circuit according to <5> or <6>, [0274] wherein each of a capacity of the third capacitor and a capacity of the fourth capacitor is variable.

<10>

[0275] The power amplifier circuit according to <2>, <4>, <5> or <6>, further comprising: [0276] a resistive element and a switch that are connected in series between the isolation port and the

ground.
<11>
[0277] A power amplifier circuit comprising: [0278] a carrier amplifier that amplifies a first signal, and that outputs a first amplified signal from a first output terminal; [0279] a peak amplifier that amplifies a second signal different in phase from the first signal, and that outputs a second amplified signal from a second output terminal; [0280] a first inductor that has a first end connected to the first output terminal, and a second end connected to a first node connected to a load; [0281] a first capacitor that has a first end connected to the first node, and a second end connected to a ground; [0282] a second inductor that has a first end connected to the second output terminal, and a second end connected to a second node; [0283] a third capacitor that has a first end connected to the second node, and a second end connected to the first node; and [0284] a fifth inductor and a second capacitor that are connected in series between the second node and the ground.
<12>
[0285] The power amplifier circuit according to <11>, further comprising: [0286] a sixth inductor that has a first end connected to the first node, and a second end connected to the load; and [0287] a fifth capacitor that has a first end connected to the first end of the sixth inductor, and a second end connected to the second end of the sixth inductor.
<13>
[0288] The power amplifier circuit according to <11> or <12>, [0289] wherein an electrostatic capacity of the second capacitor is variable.
<14>
[0290] The power amplifier circuit according to <12>, [0291] wherein an electrostatic capacity of the fifth capacitor is variable.
REFERENCE SIGNS LIST
[0292] **42** quarter-wave line [0293] **51** carrier amplifier [0294] **51***a* output terminal [0295] **52** peak amplifier [0296] **52***a* output terminal [0297] **61**, **62** LC circuit [0298] **71**, **72** 90° hybrid coupler [0299] **81** LC circuit [0300] **82** inductor [0301] **83** capacitor [0302] **86** load [0303] **86***a* antenna [0304] **101**, **102**, **111**, **112**, **122**, **132**, **141**, **900**, **901**, **902**, **903** power amplifier circuit [0305] **171**, **172**, **181** LC circuit [0306] **182** tank circuit [0307] **183***v*, **185***v*, **186** capacitor [0308] **1831**, **1832** capacitor [0309] **1833** switch [0310] **184** inductor [0311] **201**, **202**, **202***v*, **203**, **203***v*, **204**, **204***v*, **205** capacitor [0312] **2021**, **2022** capacitor [0313] **2023** switch [0314] **211**, **212**, **213**, **214**, **215**, **225** inductor [0315] **301**, **302**, **303** capacitor [0316] **311**, **312**, **313**, **314** line [0317] **315** inductor [0318] **401** switch [0319] **402** resistive element [0320] N**1**, N**2**, N**3** node [0321] Pin**1**, Pin**2** input port [0322] Pout output port [0323] Piso isolation port

Claims

1. A power amplifier circuit comprising: a carrier amplifier configured to amplify a first signal, and output a first amplified signal from a first output terminal; a peak amplifier configured to amplify a second signal different in phase from the first signal, and output a second amplified signal from a second output terminal; a first inductor having a first end connected to the first output terminal, and a second end connected to a first node; a first capacitor having a first end connected to the first node, and a second end connected to ground; a second inductor having a first end connected to the second output terminal, and a second end connected to a second node; a second capacitor having a first end connected to the second node, and a second end connected to ground; a third inductor having a first end connected to the first node, and a second end connected to a load via a third node; and a third capacitor having a first end connected to the second node, and a second end connected to the third node.
2. A power amplifier circuit comprising: a carrier amplifier configured to amplify a first signal, and

output a first amplified signal from a first output terminal; a peak amplifier configured to amplify a second signal different in phase from the first signal, and output a second amplified signal from a second output terminal; a first inductor having a first end connected to the first output terminal, and a second end connected to a first node; a first capacitor having a first end connected to the first node, and a second end connected to ground; a second inductor having a first end connected to the second output terminal, and a second end connected to a second node; a second capacitor having a first end connected to the second node, and a second end connected to ground; and a 90° hybrid coupler having a first input port, a second input port, an output port, and an isolation port, the first input port being connected to the first node and being supplied with the first amplified signal, the second input port being connected to the second node and being supplied with the second amplified signal, the output port being configured to output an output signal and connected to a load via a third node.

3. The power amplifier circuit according to claim 1, further comprising: a fourth capacitor having a first end connected to the first node, and a second end as an isolation port; and a fourth inductor having a first end connected to the second node, and a second end connected to the second end of the fourth capacitor, wherein the third inductor is electromagnetically coupled to the fourth inductor.

4. The power amplifier circuit according to claim 1, wherein a magnitude of a difference between a change in phase of the first amplified signal and a change in phase of the second amplified signal is greater than or equal to 60° and less than or equal to 120° when the first amplified signal passes through the first node, the third node, and the load, and the second amplified signal passes through the second node, the third node, and the load.

5. The power amplifier circuit according to claim 2, wherein a magnitude of a difference between a change in phase of the first amplified signal and a change in phase of the second amplified signal is greater than or equal to 60° and less than or equal to 120° when the first amplified signal passes through the first node, the third node, and the load, and the second amplified signal passes through the second node, the third node, and the load.

6. The power amplifier circuit according to claim 1, wherein a magnitude of a change in phase of the first amplified signal is greater than or equal to 45° and less than or equal to 135° when the first amplified signal passes from the first output terminal through the first node and the second node to the second output terminal.

7. The power amplifier circuit according to claim 2, wherein a magnitude of a change in phase of the first amplified signal is greater than or equal to 45° and less than or equal to 135° when the first amplified signal passes from the first output terminal through the first node and the second node to the second output terminal.

8. The power amplifier circuit according to claim 3, wherein the third capacitor and the fourth capacitor are variable capacitors.

9. The power amplifier circuit according to claim 2, further comprising: a resistive circuit element and a switch connected in series between the isolation port and the ground.

10. A power amplifier circuit comprising: a carrier amplifier configured to amplify a first signal, and output a first amplified signal from a first output terminal; a peak amplifier configured to amplify a second signal different in phase from the first signal, and output a second amplified signal from a second output terminal; a first inductor having a first end connected to the first output terminal, and a second end connected to a load via a first node; a first capacitor having a first end connected to the first node, and a second end connected to ground; a second inductor having a first end connected to the second output terminal, and a second end connected to a second node; a third capacitor having a first end connected to the second node, and a second end connected to the first node; and a fifth inductor and a second capacitor connected in series between the second node and ground.

11. The power amplifier circuit according to claim 10, further comprising: a sixth inductor having a first end connected to the first node, and a second end connected to the load; and a fifth capacitor having a first end connected to the first end of the sixth inductor, and a second end connected to the second end of the sixth inductor.

12. The power amplifier circuit according to claim 10, wherein the second capacitor is a variable capacitor.

13. The power amplifier circuit according to claim 11, wherein the fifth capacitor is a variable capacitor.